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while True:
    ret, frame = cap.read()
    if not ret:
        time.sleep(0.01)
        continue

    hsv = cv2.cvtColor(frame, cv2.COLOR_BGR2HSV)
    mask = cv2.inRange(hsv, color_lower, color_upper)
    contours, _ = cv2.findContours(mask, cv2.RETR_EXTERNAL,
cv2.CHAIN_APPROX_NONE)

    raw_dist = sonar.getDistance() / 10.0
    distance_data.append(raw_dist)
    data = pd.DataFrame(distance_data)
    data_ = data.copy()
    u = data_.mean()
    std = data_.std()

    data_c = data[np.abs(data - u) <= std]
    if len(data_c) > 0:
        distance = data_c.mean()[0]
    else:
        distance = raw_dist

    if len(distance_data) == 5:
        distance_data.remove(distance_data[0])

    cv2.rectangle(frame, (ZONE_X_MIN, ZONE_Y_MIN), (ZONE_X_MAX, ZONE_Y_MAX), (0,
255, 255), 2)

    if len(contours) > 0 and etat != "ATTRAPER":
        c = max(contours, key=cv2.contourArea)
        area = cv2.contourArea(c)

        if area > 500:
            M = cv2.moments(c)
            if M['m00'] != 0:
                cx = int(M['m10']/M['m00'])
                cy = int(M['m01']/M['m00'])

                x, y, w, h = cv2.boundingRect(c)
                cv2.rectangle(frame, (x, y), (x+w, y+h), (0, 255, 0), 2)
                cv2.circle(frame, (cx, cy), 5, (255, 0, 0), -1)

                texte_info = f"X={cx} Y={cy} Dist={distance:.1f}cm"
                cv2.putText(frame, texte_info, (10, 20),
cv2.FONT_HERSHEY_SIMPLEX, 0.5, (0, 255, 0), 2)

                centre_x_ok = ZONE_X_MIN < cx < ZONE_X_MAX
                centre_y_ok = ZONE_Y_MIN < cy < ZONE_Y_MAX

                if centre_x_ok and centre_y_ok:
                    cv2.putText(frame, "CIBLE CENTREE !", (60, 50),
cv2.FONT_HERSHEY_SIMPLEX, 0.7, (0, 0, 255), 2)

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cv2.imshow("Camera Couleur", frame)
cv2.waitKey(1)

etat = "ATTRAPER"
print(f"Cube centre sur l'image (X={cx}, Y={cy}) ! Je
m'arrete !")

chassis.set_velocity(0, 0, 0)
time.sleep(1)

avancer_et_baisser_bras()
time.sleep(1.5)

fermer_pince()
time.sleep(1)

relever_bras()
time.sleep(1.5)

print("Cube attrape ! Mission accomplie.")
break

else:
    etat = "APPROCHER"

    if cx < 250:
        faire_un_pas(0, 90, -0.1)
    elif cx > 390:
        faire_un_pas(0, 90, 0.1)

    else:
        faire_un_pas(10, 90, 0)

else:
    etat = "CHERCHER"
    faire_un_pas(0, 0, 0.2)

else:
    etat = "CHERCHER"
    faire_un_pas(0, 0, 0.2)
    cv2.putText(frame, "Recherche de la cible...", (10, 20),
cv2.FONT_HERSHEY_SIMPLEX, 0.5, (0, 0, 255), 2)

cv2.imshow("Camera Couleur", frame)
cv2.waitKey(1)

```